Every day, 2,000 tonnes of Nairobi's waste reach Dandora.

We turn it into roads that heal themselves.



Leslie Owuor · Founder, Smart Asphalt

Born and raised in Dandora · Civil Engineering, JKUAT

Photo: Dandora dumpsite, Falkue, CC BY-SA 4.0, Wikimedia Commons

IN DANDORA

Few jobs. Toxic air. A mountain of value going to waste.

KES 200

a day: as little as many waste pickers earn, in toxic smoke

1 in 2

children tested near the dumpsite had high blood lead

ACROSS KENYA

Roads keep breaking, while the waste that could help prevent them burns in Dandora.

30%

of Kenya's roads are in poor condition

100%

of the bitumen that holds them together is imported

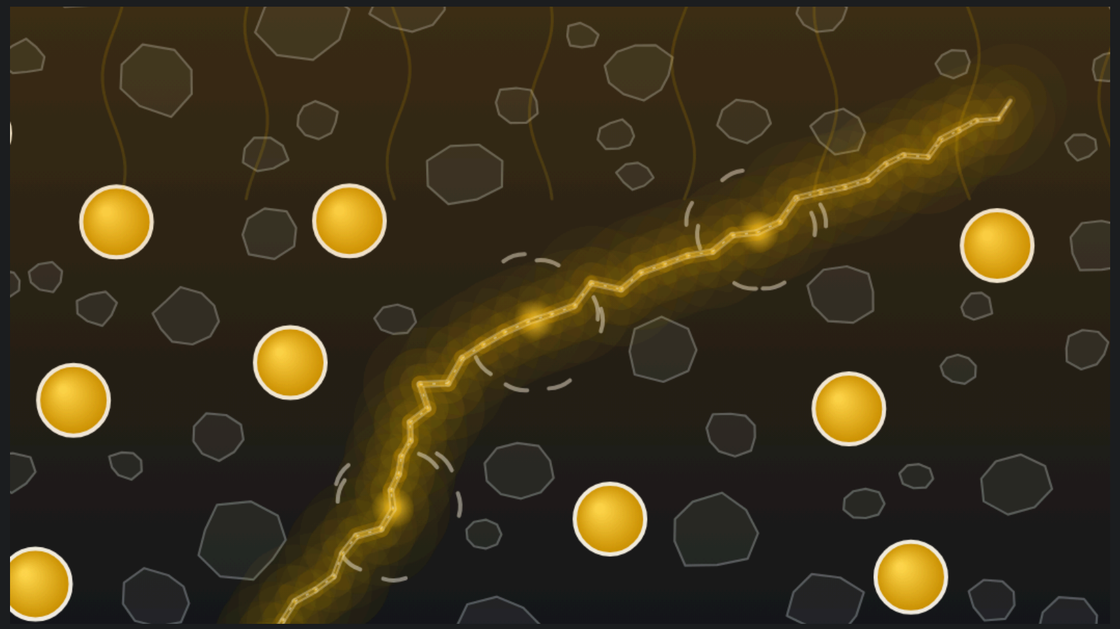
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WHAT SMART ASPHALT DOES

Roads that heal themselves, made from Dandora’s waste.



A crack starts under traffic…



…oil capsules open and the sun helps it seal.

Fewer potholes
for every road user

Lower repair bills
for road agencies: ~30% less over 20 years*

Paid work
for Dandora’s waste pickers

Illustration from our animation. *Our model, to be proven in the pilot.

PROOF, NOT THEORY

40,000 km

of road in India are already built with waste plastic.

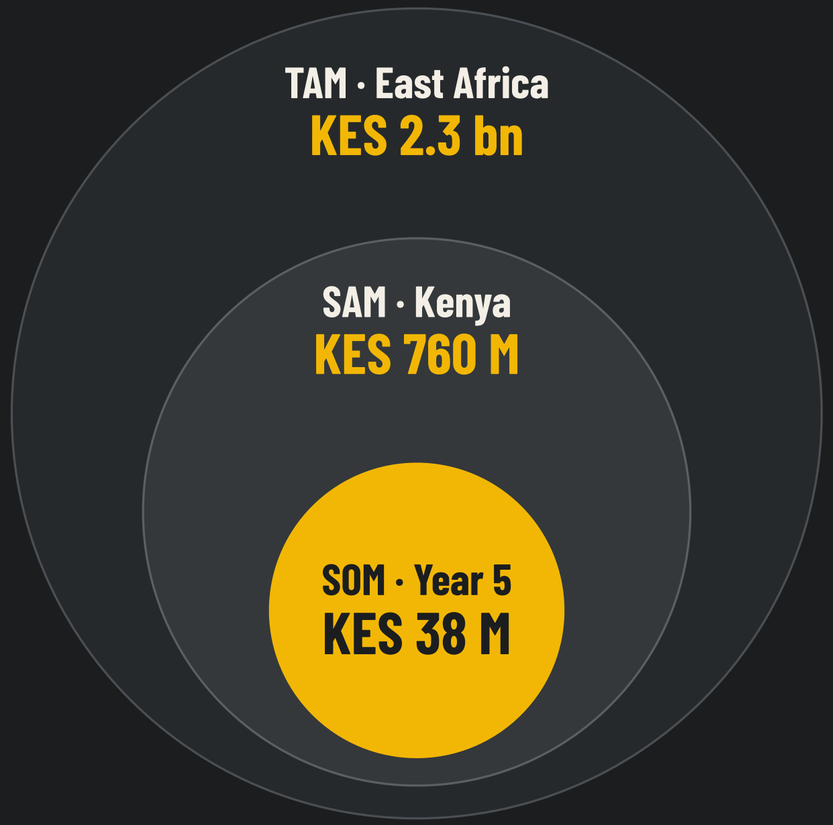
- Healing capsules survive hot asphalt production (TU Delft research)
- Used cooking oil restores old, hardened bitumen (laboratory studies)
- Our groundwork: a four-phase study, 33 claims checked against sources, a working healing simulator

Scan for our research

Full reports, sources and the healing simulator

THE MARKET

A KES 2.3 billion-a-year market, and a gap nobody fills.



Others make roads stronger. We make them heal.

Japanese PET-asphalt study in Nairobi
Nippon Road, Kao, Shimizu: stronger roads from bottle plastic

Imported polymer-modified bitumen
stronger, but costly and imported

Pothole patching
fixes damage after it happens

Our edge: self-healing, waste nobody buys, built for Kenyan heat, and a supply chain rooted in Dandora.

HOW WE MAKE MONEY

We buy from Dandora and sell to road builders.

1 Buy — Plastic film from Dandora's waste pickers, above today's price, and used cooking oil from eateries

2 Make — Our healing additive, at a small hub near the dumpsite

3 Sell — To asphalt plants and contractors, per tonne of asphalt. Later, licence the recipe

KES 700 per tonne of asphalt · about 22% margin*



Photo: Dandora dumping site, SuSanA Secretariat, CC BY 2.0. *Our model.

THE PILOT

We test it where it's hardest: under heavy trucks, in Dandora.

1 · Lab

Months 1–4

Prove capsules survive hot mixing, and measure how much cracks heal

2 · Truck yard

Months 5–8

A private truck-parking yard in Dandora: Smart Asphalt beside normal asphalt, under daily heavy trucks

3 · Jobs

From month 3

A sorting hub near the dumpsite employing 6 waste pickers

4 · Public road

Month 12+

With the yard data, a public road with KURA or KeNHA approval

Proof first, approvals next, and paid work from the start.

Photo: road beside the Dandora dumping site. SuSanA Secretariat, CC BY 2.0



THE TEAM

Leslie Owuor, Founder

- Born and raised in Dandora: I know the community and the waste pickers' reality first-hand
- Civil Engineering student at JKUAT, trained in pavements and materials
- Led the four-phase research behind Smart Asphalt

Bringing in

A materials lab partner · a road agency (KURA or KeNHA) · Dandora waste-picker groups · a business mentor · fellow JKUAT engineers

THE ASK

We're raising KES 3,000,000.

35%	Lab validation: capsule survival and healing · KES 1,050,000
25%	Truck-yard pilot in Dandora · KES 750,000
20%	Sorting hub: equipment and 6 jobs · KES 600,000
10%	Sensors and data · KES 300,000
10%	Approvals and contingency · KES 300,000

Plus mentorship, lab access, and introductions to KURA, KeNHA, KeRRA, Nairobi County and KEBS.

Dandora's waste can build Kenya's roads.

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Scan for our research



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